

Serviceability of earthquake-damaged water systems: Effects of electrical power availability and power backup systems on system vulnerability

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Abstract

Civil infrastructure systems, such as water, electrical power, natural gas, and transportation systems, are essential to the smooth functioning of modern society. Because of their inter-connected nature, once one infrastructure system is damaged by an earthquake or other natural hazard, other infrastructure systems may malfunction as well. A number of previous studies have assessed vulnerability of infrastructure systems to earthquakes, but seldom have failures due to infrastructure system interactions been considered. In its assessment of the earthquake-induced damage of a municipal water system, this paper includes the impact of damage to the supporting electrical power system using a fault tree analysis and a shortest-path algorithm. The effect of uncertainty of seismic intensity and component fragility on network integrity is evaluated. A case study involving a simple model of the electrical power system and water system in Shelby County, TN, which includes the city of Memphis, demonstrates the importance of taking infrastructure interactions into account when evaluating the seismic vulnerability and risk to a networked system, as well as the utility of back-up power systems in electric power facilities.

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1. Introduction

Commerce, industrial production and public health and safety depend strongly on the availability of distributed civil infrastructure systems, including electrical power transmission, water and gas distribution, and public transportation systems. The serviceability of such systems can be measured by the ratio of the satisfied customer demand to total customer demand within the area served. If the serviceability of these infrastructure systems is impaired or destroyed as a consequence of natural hazards or man-made accidents or incidents, business and industrial activities in the community serviced are interrupted and productivity is diminished or lost. The recovery period

usually is longer when the systems are more severely damaged, and the losses increase exponentially as the period required to fully recover serviceability of such infrastructure systems lengthens [1]. The demands placed on an infrastructure system, as well as the capacity of components and systems to withstand those demands, are highly uncertain. Accordingly, a probability-based vulnerability analysis of infrastructure systems is necessary to forecast the likely impact of a disaster on their operating requirements and the degree of damage or diminished functionality in key system components. This vulnerability assessment is one basis for estimating probable losses and the impact of diminished functionality on the affected community.

Infrastructure systems that may be impacted by a natural or man-made disaster consist of many interconnected components and delivery systems. These components are widely distributed over a region and may sustain different levels of damage when subjected to an extreme

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event. For example, a municipal water system consists of water storage tanks, pumping stations and pipelines. Electrical power systems in an urban area may have one or more generation plants, substations and transmission and distribution lines. Furthermore, civil infrastructure systems often are functionally interdependent. A recent and familiar example of the impact of functional dependence between infrastructure systems on their operating characteristics occurred during the widespread power outage in the northeastern United States and parts of Canada in August, 2003. During that outage, the water system at Cleveland, OH lost serviceability, even though the facilities within the system itself did not sustain any physical damage, because the pumping stations required electrical power to operate [2].

The impact of the serviceability of infrastructure systems, defined by the aggregate functionality of facilities composing the systems, on a community following an earthquake has received considerable recent attention. For example, Shinozuka et al. [3] analyzed the earthquake damage to a water system in Memphis, TN, and evaluated the ratio of water supply to customer demand under a scenario earthquake with an epicenter at Marked Tree, AK and moment magnitude M_w equal to 7.5. Hwang et al. [4] developed a method for evaluating seismic performance of water delivery systems, and examined the serviceability of the water system in Memphis, TN in terms of availability of both water pressure and water flow following a scenario earthquake centered at Marked Tree, AK with M_w equal to 6.5. In a similar vein, Pires et al. [5] and Ang et al. [6] evaluated the seismic performance and serviceability of the electrical power system in the San Francisco Bay area following the 1989 Loma Prieta earthquake, considering both disruptions of connection between substations and abnormal power surges due to electrical power imbalance or abnormal power voltage in power transmission lines. Shinozuka et al. [7] evaluated the serviceability of the electric power system managed by Memphis Light, Gas and Water (MLGW) Division considering loss of connectivity between substations, failure of substations and power system imbalance under a scenario earthquake centered at Marked Tree, AK with M_w equal to 7.5. Subsequently, Shinozuka et al. [8] introduced a method to evaluate the performance of electrical power systems sustaining earthquake damage that considers both disconnectivity and abnormal power flow. This method was applied to the electrical power system managed by Los Angeles Department of Water and Power (LADWP) to evaluate the electrical power availability in the Los Angeles area under the 1994 Northridge earthquake. Jin et al. [9] extended this method in developing a risk curve which displayed “the annual probability of the system’s capacity for power production decreasing to a range less than a specified level”. The studies summarized above consider the performance of specific infrastructure systems to a scenario earthquake, and focus on particular features of those systems.

None of the analyses cited above considered interdependency effects among infrastructure systems. Such interdependencies might impact the performance of individual systems significantly.

This paper proposes a method to analyze the serviceability of a municipal water system following an earthquake, taking into account (a) the dependence of the water system functionality on the availability of electrical power, and (b) the effectiveness of an auxiliary power backup system which is intended to supply temporary power to key components of the water distribution system in the event that the main power source is damaged or lost. Features of the model are illustrated through an analysis of the vulnerability of the municipal water system in Memphis, TN, following an earthquake in the NMSZ. Sources of uncertainty due to ground motion attenuation and facility performance are considered. The ground motion attenuation is modeled as a random function, leading to a spatially random seismic intensity over the affected area. The failure of the water distribution system depends on the number of pipe breaks/km of pipe, which is random, and the (uncertain) damage states (DS) of the pumping stations, storage tanks and other support facilities, which are described by their fragilities. The functionality of the water distribution system, which is modeled as consisting of facilities interconnected by distributing elements, is evaluated through a network accessibility algorithm [10,11]. This functionality, coupled with a measure of its uncertainty, can provide a basis for subsequent loss estimation and public decision making. The models of water distribution and electrical power systems have been idealized for purposes of demonstrating the feasibility of the proposed model. However, the results of the vulnerability assessment show clearly the importance of considering infrastructure system interdependencies when the serviceability of a water system is evaluated, and the enhancement to serviceability from installing a power backup system to guard against an electrical power outage caused by earthquake damage.

2. Serviceability assessment procedure

The proposed method for evaluating serviceability of a water distribution system through network accessibility analysis has five major steps:

1. The service area of the electrical power and water distribution systems must be determined and the location and function of key facilities and distributing elements of these systems within that service area must be identified. Water and power flow directions for each distributing element must be identified.
2. Seismic intensities from the selected scenario earthquake (described subsequently) at each facility (node) and distributing element within each network must be estimated. In order to treat the seismic intensity as a random variable, its mean (or median), coefficient of

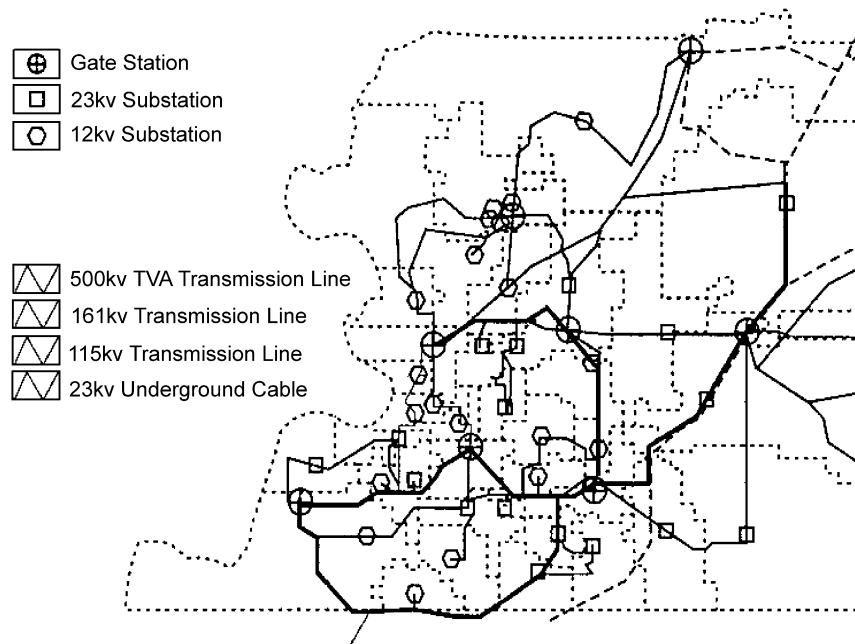


Fig. 1. Electrical power system in Shelby County, TN [7].

variation (COV) and cumulative probability distribution (CDF) must be introduced.

3. The failure probability of each water and electrical power facility and each distributing element is evaluated from an appropriate fragility curve for the facility and break density of the distributing element. The shortest-path algorithm is used to integrate the damage of facilities into the damage of the networked system.
4. Probabilistic assessment of the damaged water system is obtained by considering dependency of water facilities on electrical power.
5. The probability distribution of serviceability of the earthquake-damaged water distribution system and mean functionality of each facility is evaluated by Monte Carlo simulation. Serviceability and functionality are defined quantitatively in the following sections.

Details of the method summarized above are illustrated through a study of the water system and electrical power system in Shelby County, TN, which includes the Memphis metropolitan area. This case study is described next.

3. Description of water and power systems in Shelby County, TN

The water and electrical power systems in Shelby County, TN, which includes the city of Memphis, consist of numerous facilities and widespread distribution networks. For purposes of illustrating the potential significance of network interdependence and to avoid undue complexity, only major facilities and trunk distributing elements are considered in this study. Furthermore, since there are no electrical generating plants in Shelby County,

large substations in the metropolitan area (termed “gate stations”) provide the electric power generated by plants outside the county, and are considered as supply facilities for electrical power. Smaller electrical power substations and water distribution nodes within Shelby County are considered as demand facilities, which relay the commodity (water or electrical power) to other demand facilities and customers. Electrical transmission lines and water mains are considered as distributing elements which route the commodity from facilities to facilities and customers.

Both the water system and electrical power system are managed by the Memphis Light, Gas, and Water (MLGW) Division in Shelby County, TN. The water system consists of 6 elevated storage tanks, nine pumping stations and numerous water distribution nodes connected by buried pipe [4,12]. The electrical power system consists of 8 gate stations, sixteen 23 kV substations and 21 12 kV substations. Each substation has its own service area, within which it provides electrical power to customers [7].

Figs. 1 and 2 illustrate the main electrical power and water systems in Shelby County, TN. The service areas of the electrical power system are represented by the dashed lines in Fig. 1. These service areas are required to analyze the dependency of the water system on the electrical power system in a later section.

3.1. Scenario earthquake and seismic intensity

The New Madrid seismic zone (NMSZ), centered in Southeast Missouri, represents a major seismic hazard for several population centers in the Central and Eastern United States (CEUS). Very large earthquakes have occurred historically in this zone, such as the 1811–1812

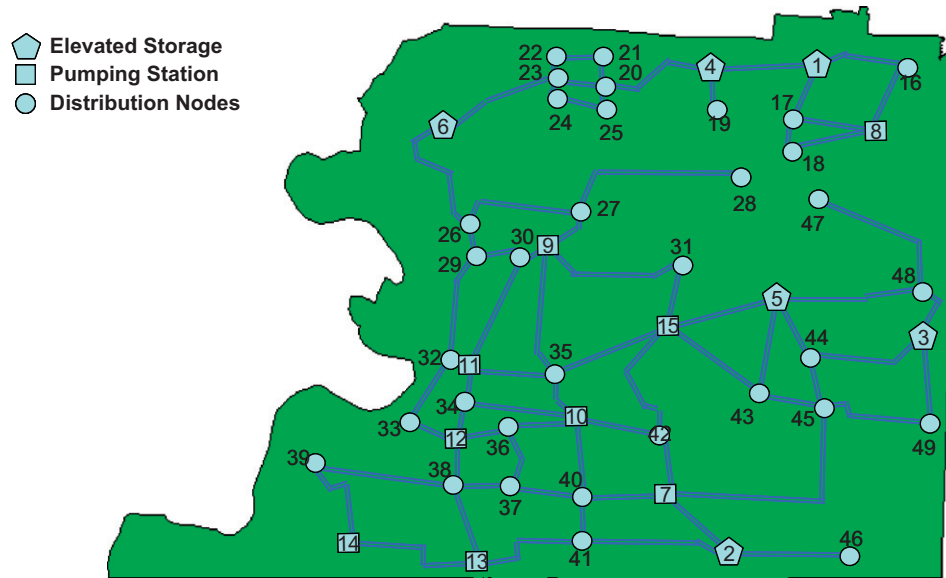


Fig. 2. Water system in Shelby County, TN (modified from Chang, S.E [13]).

sequence, and likely to occur in the future². The characteristic attenuation of strong ground motion in the CEUS is noticeably different from that in the Western United States. According to Nuttli [14], the 0.25 g peak horizontal acceleration contours for the New Madrid earthquake ($m_b = 7.2$) enclose an area that is about 11 times larger than the comparable area for a southern California earthquake of approximately the same magnitude ($M_L = 7.2$), and the area enclosed by the 20 cm/s peak horizontal ground velocity contour was 8.4 times larger. Shelby County, TN, to which the proposed method will be applied, is located approximately 20–100 km from the NMSZ and is the largest major population center potentially affected by major earthquakes in the NMSZ.

Probabilistic seismic hazard analysis (PSHA) provides estimates of mean annual rate of occurrence or annual probability that ground motions exceed a specific intensity over a range of intensities [15]. Because PSHA can be used to determine point or interval estimates of risk based on all possible earthquakes, this method has become a common seismic risk assessment tool and provides the most informative estimate of earthquake risk in a region or at a site. On the other hand, the results from a PSHA are sometimes difficult for nonspecialist decision makers, such as public officials and city planners, to interpret because the significant earthquake threats corresponding to the low probabilities of interest in safety analysis of civil infrastructure (on the order of 2% in 50 yr, or 0.0004/yr) represent an aggregation of earthquake events rather than one specific earthquake. Furthermore, this aggregated event cannot describe the spatial variability of damaging intensities across a region due to any particular severe

earthquake [16–18], and thus may not be appropriate for assessing risk to a distributed infrastructure system. A risk assessment based on a scenario earthquake avoids these difficulties, but the estimated risk is conditioned on the occurrence of the scenario event. Appropriate scenario events can be determined by a process known as disaggregation [19], which identifies the dominant seismic events contributing to an earthquake hazard of 2% in 50 yr, the current basis for building design in the United States [21,22], or some other stipulated hazard measure for the region being evaluated. The scenario earthquake used to illustrate the network vulnerability analysis in this paper is one with M_w is 7.7, with an epicenter located at 35.3N and 90.3W, which is determined from the result of the disaggregation³ analysis provided by the US Geological Survey [23]. The average epicentral distance to the city of Memphis is about 39 km. The authors believe that the threat to an urban area from such an event may be more readily appreciated by a nonspecialist decision maker than the threat from a “2% in 50 yr” earthquake.

The seismic intensity at a site from an earthquake with magnitude M_w and epicentral distance R from that site is determined by an attenuation equation, modified by the local soil conditions. For the CEUS, attenuation relations have been suggested by Frankel et al. [24], Toro et al. [25], Atkinson and Boore [26] [27], Dahle et al. [28], Hwang and Huo [29], Toro [30] and Campbell [31]. The attenuation equations describe the median ground motion intensity (defined as either peak ground or spectral parameters) as a function of M_w and R . There is considerable scatter around the estimated median, and the COV in the ground motions predicted by these attenuation curves typically are on the order of 0.60–0.80. According to Hwang [4], Shelby County is located within the Mississippi embayment, and

²Recent studies of seismicity in the New Madrid region suggest that earthquakes of such magnitudes have a return period of approximately 500 yr [20].

³<<http://earthquake.usgs.gov/>>.

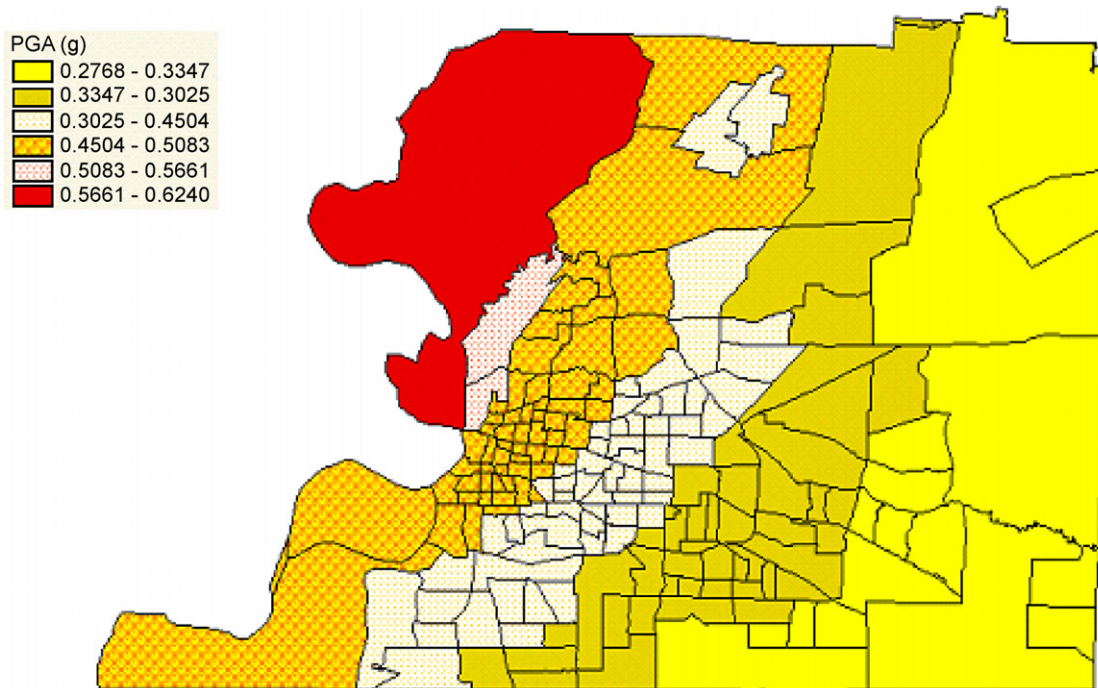


Fig. 3. Median peak ground acceleration in Shelby County (modified from HAZUS 97[32]).

the local site conditions in the region are only site categories D (stiff soils) and E (soft soils) as specified in the 2003 NEHRP provisions [21]. In this paper, the attenuation equation for PGA proposed by Toro et al. [25], which is included in HAZUS⁴ [32], is selected; the associated COV associated with this relationship, which measures the aleatoric uncertainty⁵ in ground motion intensity at the site [25], is 0.58. The soil condition is assumed to be categorized by site category D over the entire area in the absence of detailed information on local site conditions. Figs. 3 and 4 show the median peak ground accelerations (PGA) and the median peak ground velocities (PGV) produced by HAZUS for Shelby County, TN by census tract. The PGA and PGV are common measures of earthquake intensity for assessing distributed civil infrastructure damage, as described subsequently.

3.2. Seismic damage assessment of infrastructure systems

In the network algorithm utilized in this paper, *accessibility* represents the situation where a demand

facility can receive the commodity (water or electrical power) from at least one supply facility through distributing elements and other facilities.

In order to assess the accessibility of seismic-damaged infrastructure systems, two separate analyses must be conducted and properly integrated: (1) the seismic damage assessment of distributing elements, and (2) the seismic damage assessment of facilities.

3.3. Accessibility assessment of infrastructure systems

The demand facilities (water pumping stations, water distribution nodes and electrical power substations) have an important role in relaying water or electrical power from the supply facilities (water storage tanks and gate stations) to customers through water mains and electrical power transmission lines. Because of these roles, the functionality of demand facilities following an earthquake depends on not only their own physical damage but also loss of accessibility from the supply facilities. An accessibility algorithm can be used to describe the functionality of facilities which comprise a distribution network.

The Floyd–Warshall algorithm [10,11] has been widely used in the operations research field to find the shortest paths between all pairs of nodes in a network because of its effectiveness and simple programmability. When one facility cannot access another facility, the path length is set equal to infinity (in computation, a very large number), that is, the commodity is prevented from being conveyed between these facilities. The flow direction of the commodity in the distributing element is a key ingredient of accessibility assessment. Because the information available

⁴HAZUS is a risk assessment platform that was developed by the National Institute of Building Sciences, under sponsorship by the Federal Emergency Management Agency, for determining risk to communities from several natural hazards, including earthquakes.

⁵The aleatoric uncertainty (inherent randomness) describes the variability in ground motion intensity with respect to the attenuation law due to uncertainties in the source, source-to-site, and local site effects associated with a stipulated seismogenic zone. In contrast, the epistemic (knowledge-based) uncertainty represents the uncertainty resulting from alternative seismic zone hypotheses, as well as from the attenuation law selected to determine the site intensity.

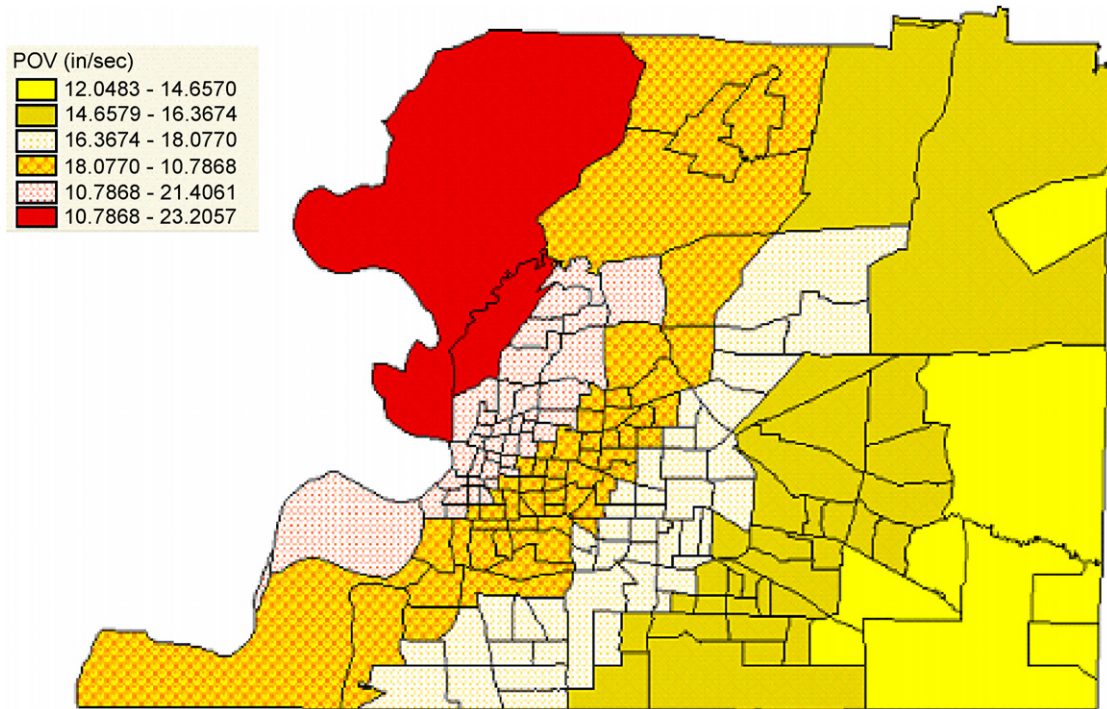


Fig. 4. Median peak ground velocity in Shelby County (modified from HAZUS 97 [32]).

in this preliminary investigation is limited, the following assumptions are adopted for water flow and electrical power flow: (1) The supply facilities can convey the commodity to any other facility as long as they are directly connected by distributing elements, and (2) the demand facilities can convey the commodity to other substations, water pumping stations and water distribution nodes as long as they are directly connected, but cannot convey the commodity to gate stations and water storage tanks.

3.4. Damage assessment of distributing elements

Water and electrical power are delivered from supply facilities to customers through demand facilities and distributing elements. It is assumed that damage to electrical power distribution systems can be assessed from the damage to transmission towers since the transmission lines, which are suspended from the transmission towers, are highly compliant and thus are essentially free from seismic excitation [9]. On the other hand, the distributing elements in the water system can be assessed from the damage to water mains. Previous research analyzing the seismic vulnerability of water piping has led to relations between seismic intensity and pipe-break density [33–35]. The *Guidelines* prepared by the American Lifeline Alliance (ALA) [38] suggests that damage to water pipe caused by strong ground motion can be expressed as a function of PGV:

$$RR = K(0.00187)PGV, \quad (1)$$

where RR is the repair ratio, which is the number of pipe breaks/1000 ft (305 m) of pipe length, K is a coefficient

determined by the pipe material, pipe joint type, pipe diameter and soil condition, and PGV has the units of in/s. Water mains installed in the Memphis area before 1960 were unlined cast iron; those installed between 1960 and 1975 were cement-lined cast iron and ductile iron [3], and those installed after 1975 were ductile iron [4]. Only large-diameter water mains whose diameters are from 6 to 48 in (152–1220 mm) [3,4] are considered in this study. In view of the situation at Memphis, it is assumed that $K = 0.5$ (ALA, 2001). According to Broughton et al. [36], the flood plains along the Wolf, Loosahatchie and Mississippi Rivers in Shelby County may be susceptible to liquefaction. However, Broughton's liquefaction susceptibility map [36] shows that zones with very high liquefaction susceptibility cover a relatively small area of Shelby County. Furthermore, liquefaction data from the Northridge earthquake [37] indicated that permanent ground displacements (PGD) due to liquefaction occurred only in zones where the PGV exceeded 30 in/s. (75 cm/s), a value that is much larger than the attenuated median PGV in Shelby county shown in Fig. 4. Thus, the possibility of pipe break caused by PGD due to liquefaction [38–40] is neglected.

Under the assumption that the seismic intensity leads to a uniform demand on a water pipe connecting two facilities, the number of pipe breaks can be expressed by the Poisson probability law:

$$P[N = n] = e^{-RR L} \cdot \frac{(RR L)^n}{n!}, \quad (2)$$

where n is the number of pipe breaks, RR the repair ratio evaluated by Eq. (1), and L the length of pipe segment (expressed in terms of 1000-ft segments). If, in addition, it

is assumed that a pipe segment cannot deliver water when the segment has at least one pipe break, the failure probability of the pipe segment can be expressed by the exponential distribution:

$$P_f = 1 - P[N = 0] = 1 - e^{-RR L} \quad (3)$$

In this study, the events describing failure of each pipe segment are assumed to be statistically independent.

3.5. Damage assessment of facilities

The functionality of a municipal water system following an earthquake is determined by not only the physical damage to the system itself but also the serviceability of electrical power systems to which it is connected. Furthermore, some facilities have backup systems to provide electrical power in the event that power from the primary electrical grid is lost or impaired; in this case, the effect of any backup system also should be considered. For example, some water pumping stations have auxiliary electric power generators [2].

The simple fault tree model in Fig. 5 illustrates how the functionalities of the primary and supporting systems are integrated in the overall system analysis.

Using this fault tree model and assuming, for simplicity, that a facility is either fully functional or nonfunctional, the probability of lost functionality of an earthquake-damaged water facility can be presented as

$$P_f = P[F_{\text{Physical}} \cup (F_{\text{Power}} \cap F_{\text{Backup}})] \quad (4)$$

where, P_f is the probability of lost functionality of a water facility, F_{Physical} the event that the water facility loses its function due to physical damage, F_{Power} the event that the necessary electrical power is not supplied to the water facility due to the lost function of an electrical power substation whose service area includes this water facility, and F_{Backup} the event that the electrical power backup system fails to function and cannot supply electrical power to the water facility.

The seismic intensities affecting a facility and its backup system at a specific site are assumed to be perfectly correlated due to their proximity, and thus are evaluated at the same seismic intensity. In contrast, intensities affecting facilities and backup systems at different sites are assumed to be

Table 1
Parameters of fragility curves (DS.V)

	Median (g)	Beta
Gate station	0.47	0.40
23 km substation	0.70	0.40
12 kV substation	0.90	0.45
Elevated water storage tank	1.50	0.60
Pumping station	1.50	0.80

stochastically independent. It is assumed that only water pumping stations have backup electrical power facilities.

The physical damage to facilities are modeled probabilistically by the seismic fragilities that are presented in the HAZUS Technical Manual [32] for various lifeline facilities, all of which are modeled by log-normal distribution functions that describe the probabilities of entering five DS, ranging from I (minor) to V (complete), as functions of PGA. We will assume herein that a facility loses its functionality completely when it is in HAZUS DS V (complete); otherwise it maintains its full functionality. As an illustration, Table 1 summarizes the parameters of the fragility curves for facilities in the water and electrical power systems having DS V. The water distribution nodes in the water distribution system (shown in Fig. 2) are assumed to be undamaged since they are simply the branch points in the piping system needed to distribute water to customers.

3.6. Serviceability assessment of infrastructure systems

In most cases, customers in a particular service area receive water or power from a facility serving that area. Thus, the serviceability of the water distribution system following the earthquake is directly related to the number of water distribution nodes in the network which remain accessible from at least one supply facility following the earthquake. In a similar way, the serviceability of the power system following the earthquake is determined from the number of substations which are accessible from at least one gate station. In this case, a measure of the serviceability ratio can be computed as

$$P[S \leq s] = P\left[S \leq \frac{\sum_{i=1}^N (\omega_i X_i)}{\sum_{i=1}^N \omega_i}\right] \quad (5)$$

where S is the serviceability ratio of an infrastructure system defined on the domain $[0,1]$, where $s = 0$ implies complete loss of serviceability, while $s = 1$ implies unimpaired function, ω_i is a weighting factor assigned to facility i , and X_i represents the functionality of facility i , which is modeled as the outcome of a Bernoulli trial ($X_i = 1$ if facility i is accessible from at least one supply facility, otherwise $X_i = 0$). N is the number of water distribution nodes or number of substations. The weighting factor ω_i is dependent on the goal of the serviceability assessment. For example, if we focus on the number of customers who can

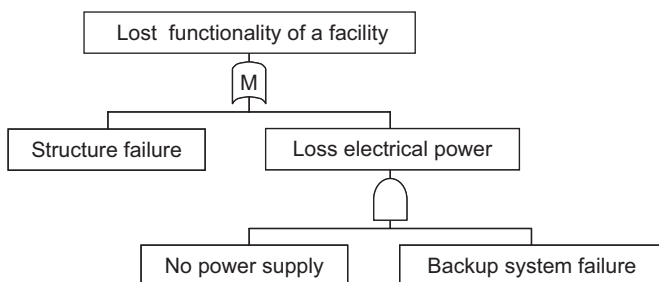


Fig. 5. Fault tree model of lost functionality of a facility.

receive the resources after an earthquake, ω_i might be determined by the number of customers in the service area of facility i . Similarly, the amount of sales in the service area of facility i can determine the weighting factor when industrial productivity is of concern. If the importance level of all demand nodes is same, Eq (5) can be rewritten as

$$P[S \leq s] = P\left[S \leq \frac{\sum_{i=1}^N X_i}{N}\right]. \quad (6)$$

In this paper, the serviceability of a water system is determined from the functionality X_i of the distribution nodes. Furthermore, it is assumed that all nodes in the water distribution system and all substations in the electrical power system have the same importance level for customers. The serviceability of the infrastructure systems is determined from Eq (6). The mean functionality ratio of water facility i is defined as the average of X_i .

3.7. Analysis of network vulnerability

The failures of facilities and the failures of distributing elements comprising an infrastructure system are integrated using the shortest-path algorithm described above [10,11]. The uncertainties due to the randomness in seismic intensity and facility response in the network analysis are propagated using Monte Carlo simulation. The process consists of the following steps:

1. Model the water and electrical power systems, which consist of facilities and distributing elements, in their original undamaged states;
2. Identify the failed distribution line(s) and remove them from the nondamaged network;

3. Identify the failed facilities and remove all distribution lines connecting to the failed facilities, yielding the damaged network system;
4. Identify inaccessible demand facilities by applying the Floyd–Warshall algorithm to the damaged network system, and assign $X_i = 0$ for inaccessible water distribution nodes and $X_i = 1$ for accessible water distribution nodes;
5. Through iteration of steps 1–4, evaluate the probability distribution of the serviceability ratio by Eq. (6) and the mean functionality ratio of facilities.

In this paper, the number of Monte Carlo trials is 10,000. Since general infrastructure systems are comprised of a very large number of facilities and distributing elements, the short periodicity of certain pseudo-random number generating algorithms may be of concern. Accordingly, pseudo-random numbers in this analysis are generated using the Mersenne Twister algorithm [41] which has long periodicity.

3.8. Serviceability of water distribution system

The serviceability assessment of the water distribution system in Shelby County, TN following the occurrence of the scenario earthquake described above leads to a set of probability distributions of serviceability ratio of the water system under various conditions of facility functionality. These distributions are illustrated in Figs. 6 and 7, and are discussed in more detail in the following paragraphs.

Fig. 6 illustrates the residual serviceability for three cases: (1) Water main failure and facility failure both are considered and are integrated (proposed method in this paper), (2) Water mains are assumed to never fail ($K = 0$ in

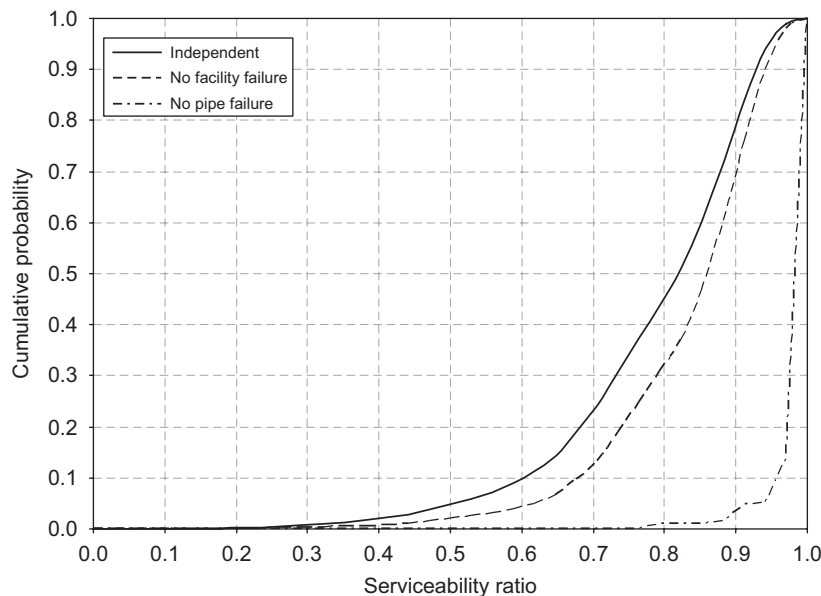


Fig. 6. Probability distribution of serviceability ratio ($P[S \leq s]$) of water system.

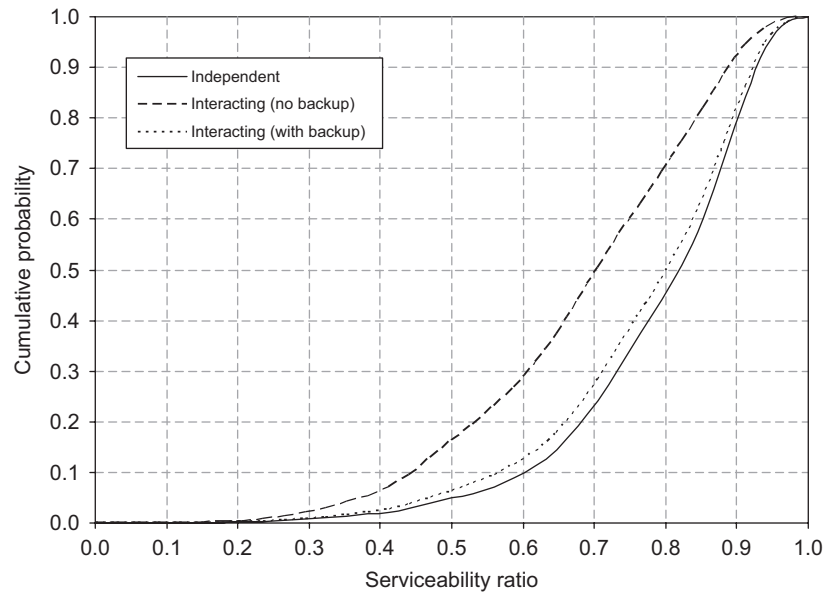


Fig. 7. The effect of water system dependency on electrical power.

Eqs. (1) and (3) Facilities are assumed to never fail ($P_f = 0$ in Eq. (4)). For example, the probability that the serviceability ratio of the water system is less than 0.8 is 5% when water main segments are assumed to never fail, 54% when facilities are assumed to never fail, and 72% when both are assumed to be susceptible to failure. These results show that the residual serviceability of a water distribution system in which failures of facilities and connecting mains both are considered is quite different from the other two cases. Thus, when the serviceability of the water system is assessed, both facility failure and water main failures should be considered simultaneously and integrated properly.

Fig. 7 illustrates the results of three additional cases of serviceability analysis: (1) Functionality of the water distribution system is not dependent on electrical power availability ($F_{\text{power}} = \text{null event}$ in Eq. (4)); (2) Functionality of the water distribution system is dependent on electrical power availability but backup electrical power sources are not installed or are unavailable ($F_{\text{Backup}} = \text{sample space}$ in Eq. (4)); and (3) Functionality of the water distribution system is dependent on electrical power availability and backup power systems are installed. The dependence of the water distribution system function on availability of electrical power is significant. Thus, the availability of electrical power is clearly an important factor in assessing water system serviceability following the occurrence of a severe earthquake. Fig. 7 shows that when power backup systems are installed in water pumping stations, the serviceability of the water system is improved; however, the effect of the interdependence between the systems remains apparent.

The mean functionality ratios of water pumping stations and water distribution nodes are plotted in Fig. 8 for the

three cases considered in Fig. 7. Items labeled 7–15 in Fig. 8 represent water pumping stations, while items identified as 16–49 represent water distribution nodes, as shown in Fig. 2. According to Fig. 8, all pumping stations show an increase in their mean functionality that approaches that of the independent case after power backup systems are installed. Following this improvement, the functionality of the water distribution nodes is also recovered. Those water distribution nodes that are not affected by power dependency and power backup installation (e.g., items 19–25 and 44–49) are nodes for which no water pumping station exists between the distribution nodes and water storage tanks. Fig. 8 also shows that the functionality ratio of water distribution nodes ranges widely, from 0.23 (item 28) to 0.97 (items 44 and 45). This range of values can be attributed to their relative distances from water storage tanks which supply water to the node, the damage of such water storage tanks, and the redundancy of the earthquake-damaged water system. Conversely, if the demand facilities are located far from supply facilities (e.g., items 14 and 39) or if their redundancy of accessibility is poor (e.g., items 28 and 47), their residual serviceability becomes small. The synergistic effect of these trends is apparent for the residual accessibility of item 28.

The impact of power backup systems on serviceability of the water system is very noticeable. Figs. 7 and 8 show that the serviceability of the water system and the functionality of the facilities comprising it are much improved if a source of electrical power can be assured. The availability of backup electrical power systems following a large earthquake is important to maintain the functionality of the water system in this case study.

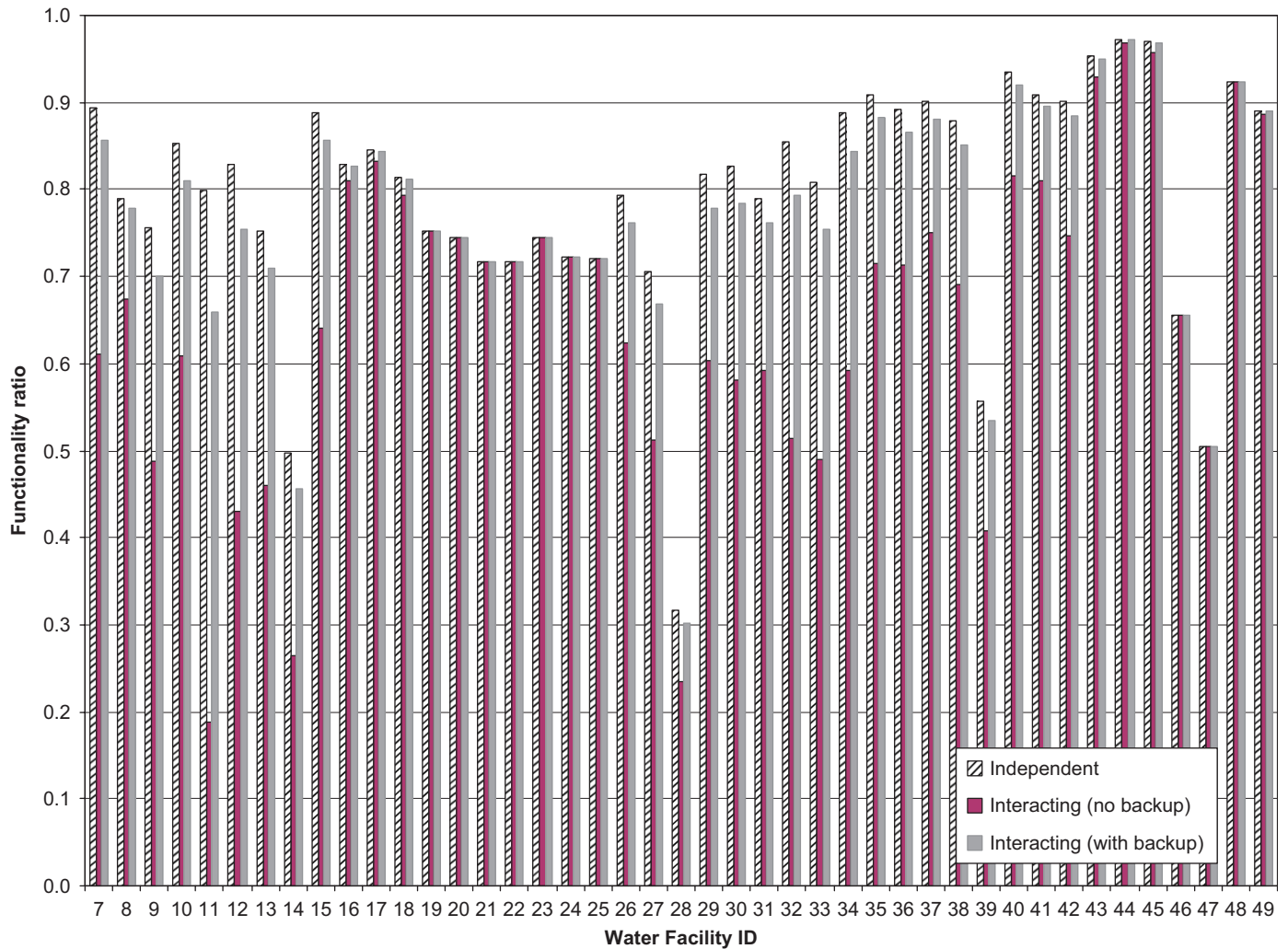


Fig. 8. Mean functionality ratio of water facility.

4. Conclusions

A methodology for evaluating the serviceability of an urban water distribution system following a scenario earthquake was presented in this paper. The dependence of water availability on the serviceability of the electrical power system and the presence of electrical power backup systems was considered. The feasibility of this method was demonstrated by an application to the water distribution and electrical power systems in Shelby County, TN. This analysis required a number of assumptions regarding soil condition, seismic intensity for water main segments, definition of pipe failure, spatial correlation of seismic intensity, water and electrical power flow direction and relation between serviceability and damage state of facility. It is believed that this method can be applied to more complicated lifeline systems, provided that the necessary supporting data can be developed.

The case study revealed that the serviceability of the water distribution system is quite different from the serviceability determined by evaluating only water main

failures or facility failures within the water system. Furthermore, the dependence of water system availability on electrical power from either/or main transmission and distribution or on-site power backup systems is significant. Thus, an accurate appraisal of the serviceability of the water system can be obtained only through a model in which an integration of water facility and water main failures is performed and in which the dependence of the water system performance on the availability of electrical power is included. Furthermore, the case study disclosed that power backup enhances the serviceability of the water distribution system significantly. Thus, the availability of a power backup system can reduce the effect of interdependency due to unavailability of electrical power from the main distribution grid.

In the case study herein, the possibility that a damaged facility can continue to function at a reduced level of efficiency was not considered. Damage to the electrical power distribution system (transmission towers and transmission lines) was neglected. Epistemic uncertainties in response of facilities and distributing elements and in

strong ground motion were not modeled; nor was any covariance in seismic intensity that may exist at adjacent points within the network considered. Research in progress is extending the model to determine the potential impact of these additional factors on network serviceability and on the mitigation of seismic risk to civil infrastructure.

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